```
%  MATLAB code to find CIRCULAR CONVOLUTION

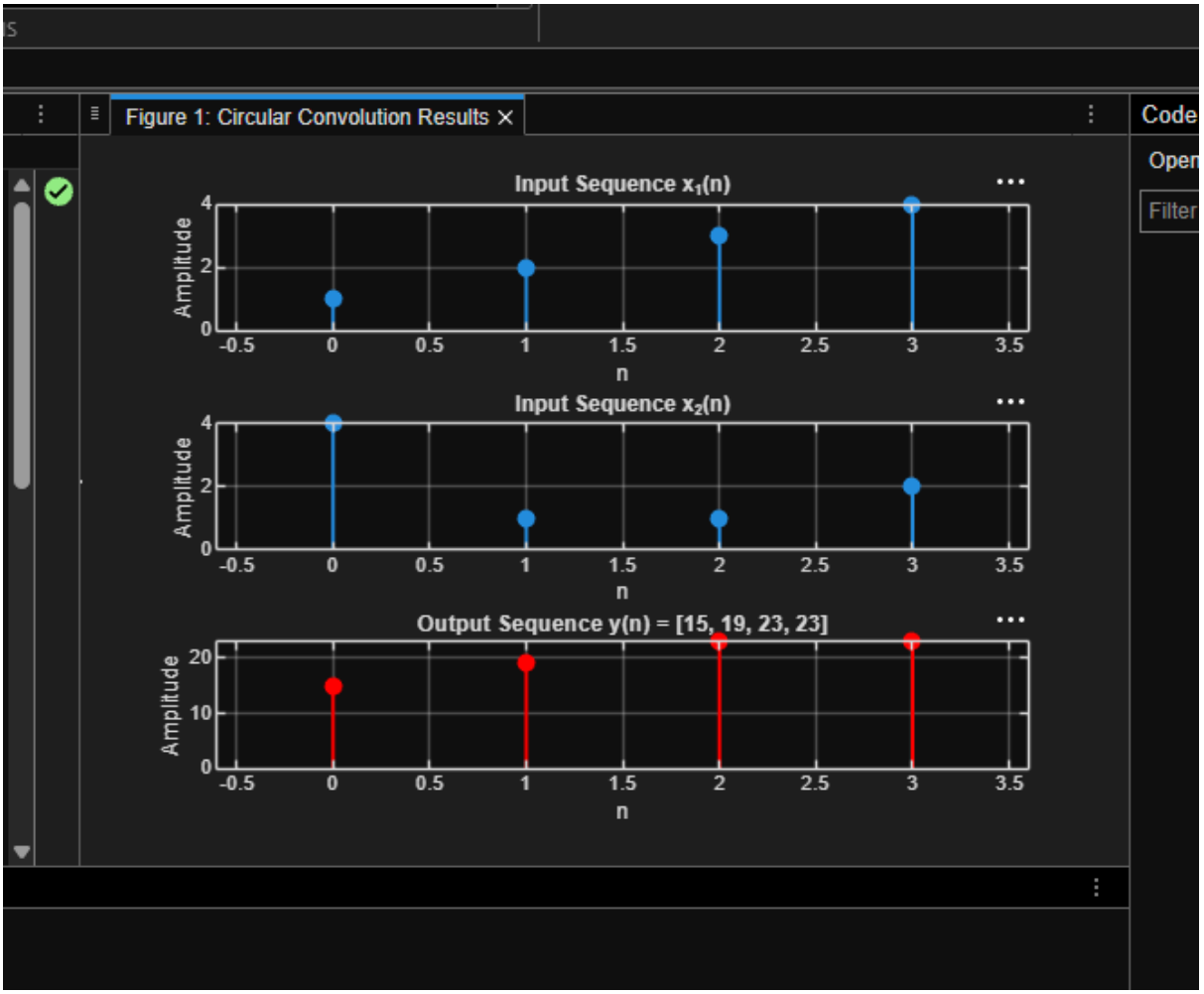
clc; clear; close all;

x1 =[1,2,3,4];
x2 =[4,1,1,2];
n = 0:3;
y = cconv(x1, x2, 4);

figure('Name', 'Circular Convolution Results');
subplot(3, 1, 1);
stem(n, x1, 'filled', 'LineWidth', 1.5);
title('Input Sequence x_1(n)');
xlabel('n'); ylabel('Amplitude'); grid on;

subplot(3, 1, 2);
stem(n, x2, 'filled', 'LineWidth', 1.5);
title('Input Sequence x_2(n)');
xlabel('n'); ylabel('Amplitude'); grid on;

subplot(3, 1, 3);
stem(n, y, 'filled', 'r', 'LineWidth', 1.5);
title('Output Sequence y(n) = [15, 19, 23, 23]');
xlabel('n'); ylabel('Amplitude'); grid on;
```


Figure 1: Circular Convolution Results
Input Sequence $x_1(n)$
Amplitude
n
Input Sequence $x_2(n)$
Amplitude
n
Output Sequence y(n) = [15, 19, 23, 23]
Amplitude
n
Code
Open
Filter

```
%  MATLAB code to find LINEAR CONVOLUTION
clear; clc;
x1 = [1, 2, 3, 4];
x2 = [4, 1, 1, 2];
y = conv(x1, x2);

disp('Linear Convolution Result (y):');
disp(y);
figure;

subplot(3,1,1); stem(x1, 'filled'); title('Sequence x1(n)'); grid on;
subplot(3,1,2); stem(x2, 'filled'); title('Sequence x2(n)'); grid on;
subplot(3,1,3); stem(y, 'filled'); title('Convolved Sequence y(n)');
 grid on;
```

